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The total market value of real estate in Altonville has steadily declined over the past four years. This decline has meant the overall figure on which the city's property tax is based—the assessed value of that real estate—has also declined. Moreover, the percentage of assessed value that was paid as property taxes to the city did not change during this period.

The information above most strongly supports which of the following?

A. Money collected from property taxes provided most of Altonville's revenue during the past four years.

B. The percentage of Altonville's overall revenue that was collected from property taxes did not change over the past four years.

C. Altonville officials had proposed raising property tax rates during the past four years but never did so because of strong opposition from property owners.

D. The total amount of revenue that Altonville has collected from property taxes has declined over the past four years.

E. During the past four years, Altonville officials also did not increase tax rates on other sources of revenue such as retail sales or business profits.

Argument Construction

Situation Altonville's real estate has declined in value in recent years, and therefore the total value of the property on which the city can charge property taxes is also reduced. The percentage of assessed value charged as property tax has remained constant.

Reasoning *What would follow from the facts presented?* A constant percentage of a total that is decreasing would likewise decrease—for example, 10 percent of eighty is less than 10 percent of one hundred. Therefore, the information supports the conclusion that Altonville's total revenue from property taxes has fallen over the period specified.

A. The passage suggests nothing about property taxes relative to other sources of revenue.

B. The passage does not suggest property taxes are unchanged as a share of Altonville's total revenue.

C. There is no mention of any proposal to raise property tax rates.

D. **Correct.** The passage states total property values have fallen while the percentage of that value levied as tax is constant. Therefore, the information supports the conclusion that revenues from property tax have fallen in tandem with the declining value of property in the city.

E. The passage implies nothing about whether taxes other than property taxes have increased.

The correct answer is D.

•

Certain groups of Asian snails include both "left-handed" and "right-handed" species, with shells coiling to the left and right, respectively. Some left-handed species have evolved from right-handed ones. Also, researchers found that snail-eating snakes in the same habitat have asymmetrical jaws, allowing them to grasp right-handed snail shells more easily. If these snakes ate more right-handed snails over time, this would have given left-handed snails an evolutionary advantage over right-handed snails, with the left-handed snails eventually becoming a new species. Thus, the snakes' asymmetrical jaws probably helped drive the emergence of the left-handed snail species.

Which of the following would, if true, most strengthen the argument that asymmetrical snake jaws helped drive left-handed snail evolution?

A. In one snake species, the snakes with asymmetrical jaws eat snails, while the snakes with symmetrical jaws do not eat snails.

B. Some species of Asian snails contain either all right-handed snails, or all left-handed snails.

C. Anatomical differences prevent left-handed snails from mating easily with right-handed snails.

D. Some right-handed snails in this habitat have shells with a very narrow opening that helps prevent snakes from extracting the snails from inside their shells.

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E. Experiments show that the snail-eating snakes in this habitat fail more often in trying to eat left-handed snails than in trying to eat right-handed snails.

Argument Evaluation

Situation There are both Asian snails with shells that coil to the right and Asian snails with shells that coil to the left, and the latter have evolved from the former; furthermore, there are snakes that have asymmetrical jaws that allow the snakes to grasp the snails with right-coiled shells more easily.

Reasoning *What fact would help support the claim that the snakes' asymmetrical jaws helped drive the emergence of species of snails with left-coiled shells?* We are told that if over time the snakes with asymmetrical jaws ate more snails with right-coiled shells than snails with left-coiled shells, then this would give snails with left-coiled shells an evolutionary survival advantage. So, if some evidence showed that the snakes with such jaws in fact were more likely to have successfully eaten the snails with the right-coiled shells, then we would have good reason to think the snakes' asymmetrical jaws helped drive the emergence of snails with left-coiled shells.

A. The fact that snakes with asymmetrical jaws eat snails and other snakes do not does not give any indication as to whether snails with left-coiled shells have any evolutionary advantages over snails with right-coiled shells. At best it only tells us that if snakes drove the evolution of snails with left-coiled shells, it would likely have been the snakes with asymmetrical jaws.

B. The fact that some species of Asian snails have no variation in the direction in which their shells coil tells us nothing about what, if any, evolutionary advantages they have relative to other snails, or whether the snakes' asymmetrical jaws had any effect on any such evolutionary advantages.

C. The inability of snails with left-coiled shells to mate easily with snails with right-coiled shells in and of itself tells us nothing about whether the snakes' asymmetrical jaws had any effect on the emergence of snails with left-coiled shells. Although it does suggest a different factor that could have contributed to the emergence of the snail species with left-coiled shells, it does not exclude the possibility that the snakes were also a factor.

D. The fact that snakes cannot extract some snails with right-coiled shells from their shells would suggest that these snails might have an evolutionary advantage, but the argument's conclusion is about an evolutionary advantage that snails with left-coiled shells presumably have, not an advantage that snails with right-coiled shells would have.

E. **Correct.** The fact that experiments show that the snakes are more successful at eating snails with right-coiled shells than they are at eating snails with left-coiled shells would support the claim that these snakes did in fact eat more snails with right-coiled shells, and hence the snails with left-coiled shells would as a result have an evolutionary advantage.

The correct answer is E.

•

A moderately large city is redesigning its central downtown area and is considering a plan that would reduce the number of lanes for automobiles and trucks and increase those for bicycles and pedestrians. The intent is to attract more workers and shoppers to downtown businesses by making downtown easier to reach and more pleasant to move around in.

Which of the following would, if true, most strongly support the prediction that the plan would achieve its goal?

A. People who make a habit of walking or bicycling whenever feasible derive significant health benefits from doing so.

B. Most people who prefer to shop at suburban malls instead of downtown urban areas do so because parking is easier and cheaper at the former.

C. In other moderately sized cities where measures were taken to make downtowns more accessible for walkers and cyclists, downtown businesses began to thrive.

D. If the proposed lane restrictions on drivers are rigorously enforced, more people will likely be attracted to downtown businesses than would otherwise be.

E. Most people who own and frequently ride bicycles for recreational purposes live at a significant distance from downtown urban areas.

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downtown urban areas.

Evaluation of a Plan

Situation A moderately large city desires to attract more workers and shoppers to downtown businesses by making the downtown area more pleasant and easier to travel to. In light of this goal, the city is considering a plan that would reduce the number of lanes for automobiles and trucks and increase those for bicycles and pedestrians.

Reasoning *What would most strongly support the prediction that reducing the number of lanes for automobiles and trucks and increasing those for bicycles and pedestrians would achieve the goal of attracting more people to downtown businesses?* If other moderately sized cities that have improved the accessibility of their downtown areas to pedestrians and cyclists have seen increased downtown business, then it is reasonable to think that the city in question will have improved downtown business as well.

A. If people who make a habit of walking or bicycling derive significant health benefits from doing so, that would provide further motivation for trying to encourage people to walk or ride a bicycle, but it does not provide much support for the claim that adding lanes for pedestrians and cyclists will successfully bring people to shop or work more in the city's downtown. Perhaps people will merely use the new lanes for exercise but not to get to the downtown area to do business.

B. Because the plan as described does not provide for additional parking, the fact that parking is a major concern motivating people to shop at suburban malls does not give us good reason to think that the city's plan will help achieve the city's goal.

C. **Correct.** If other moderately sized cities that have made their downtown areas more accessible to pedestrians and cyclists have seen their downtown businesses begin to thrive soon afterward, this is evidence—even if not conclusive—that the changes produced the thriving. Consequently, it is reasonable to think that the same will result for the city in question.

D. It might be the case that rigorously enforcing lane restrictions will attract more people to downtown businesses than would otherwise be the case, but the information provided does not indicate how strictly lane restrictions will be enforced.

E. The city's plan is aimed at increasing the use of bicycles for the purposes of shopping or getting to work, not only for recreational purposes. And even if people who use their bicycles for recreational purposes would also be likely to use them for shopping or getting to work, the fact that most recreational cyclists live at a significant distance from the city's downtown area may suggest that the city's plan will not be successful, because the cyclists may live too far away to use their bicycles for these purposes, even if it would be somewhat easier for them to do so once the plan has been implemented.

The correct answer is C.

•

Previously, Autoco designed all of its cars itself and then contracted with specialized parts suppliers to build parts according to its specifications. Now it plans to include its suppliers in designing the parts they are to build. Since many parts suppliers have more designers with specialized experience than Autoco has, Autoco expects this shift to reduce the overall time and cost of the design of its next new car.

Which of the following, if true, most strongly supports Autoco's expectation?

A. When suppliers provide their own designs, Autoco often needs to modify its overall design.

B. In order to provide designs for Autoco, several of the parts suppliers will have to add to their existing staffs of designers.

C. Parts and services provided by outside suppliers account for more than 50 percent of Autoco's total costs.

D. When suppliers built parts according to specifications provided by Autoco, the suppliers competed to win contracts.

E. Most of Autoco's suppliers have on hand a wide range of previously prepared parts designs that can readily be modified for a new car.

Evaluation of a Plan

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Evaluation of a Plan

Situation A car manufacturer plans to have its parts suppliers start helping to design the parts they build for the manufacturer. Many parts suppliers have more designers with specialized experience than the manufacturer has.

Reasoning *What would make it more likely that having the parts suppliers help design the parts will reduce the time and cost of designing the manufacturer's next new car?* In order for the change to reduce the time and cost, the parts suppliers involved in designing the next car will probably have to do their portion of the design process faster and cheaper than the manufacturer would have, and the design collaboration process will have to avoid producing substantial new inefficiencies.

A. The additional need to modify the overall design would probably make the design process slower and more expensive, not faster and cheaper.

B. The additional need to hire more designers would probably increase design costs, not reduce them.

C. Although this suggests that the change is likely to substantially affect the design's expense, it does not indicate whether the expense will increase or decrease.

D. If anything, this competition probably made Autoco's previous design process cheaper. It does not suggest that the new design process, which may involve less competition, will be faster or cheaper than the previous one.

E. **Correct.** Modifying the previously prepared parts designs will probably be faster and cheaper than creating new designs from scratch.

The correct answer is E.

•

In response to viral infection, the immune systems of mice typically produce antibodies that destroy the virus by binding to proteins on its surface. Mice infected with the herpesvirus generally develop keratitis, a degenerative disease affecting part of the eye. Since proteins on the surface of cells in this part of the eye closely resemble those on the herpesvirus surface, scientists hypothesize that these cases of keratitis are caused by antibodies to the herpesvirus.

Which of the following, if true, most helps to support the scientists' reasoning?

A. Other types of viruses have surface proteins that closely resemble proteins found in various organs of mice.

B. Mice that are infected with the herpesvirus but do not develop keratitis produce as many antibodies as infected mice that do develop keratitis.

C. Mice infected with a new strain of the herpesvirus that has different surface proteins did not develop keratitis.

D. Mice that have never been infected with the herpesvirus can sometimes develop keratitis.

E. There are mice that are unable to form antibodies in response to herpes infections, and these mice contract herpes at roughly the same rate as other mice.

Argument Evaluation

Situation Mice infected with the herpesvirus tend to develop keratitis, an eye disease. The surface of the eye cells has proteins that resemble those on the herpesvirus surface. Based on this finding, scientists have hypothesized that keratitis develops in mice because antibodies that attack herpesvirus surface proteins can also attack eyes.

Reasoning *What other information, if correct, would provide the strongest support for the scientists' hypothesis?* The clue that led the scientists to form their hypothesis was the close resemblance of the proteins on the mouse eye surface to those on the herpesvirus surface. The resemblance could cause antibodies to bind to both types of proteins, in one case eliminating the herpesvirus and in the other case causing keratitis.

A. Even if this is correct, we lack information as to whether the antibodies to those other types of virus can damage the organs that display the closely resembling proteins. If such a damage process were confirmed, it could count as evidence—even if not sufficient—to confirm the scientists' hypothesis.

B. If anything, this would, absent further information, raise doubts about the correctness of the scientists' proposed

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B. If anything, this would, absent further information, raise doubts about the correctness of the scientists' proposed explanation.

C. **Correct.** This provides strong confirmation of the scientists' hypothesis. The proteins on the new strain of the herpesvirus no longer sufficiently resemble the proteins on the eye surface to cause the antibodies to attack those proteins and cause keratitis.

D. For all we know, keratitis may have multiple independent causes and may sometimes be caused by processes other than the protein misidentification hypothesized by the scientists. This information neither confirms nor refutes the scientists' hypothesis.

E. The rates at which mice contract herpes is not discussed. We lack any information as to whether mice that lack antibodies to the herpesvirus sometimes contract keratitis along with herpes infection.

The correct answer is C.

•

One might expect that within a particular species, any individuals that managed to slow down the aging process would leave more offspring. Natural selection should therefore favor extreme longevity—but this does not seem to be the case. A possible explanation is that aging is a product of the inevitable wear and tear of living, similar to how household appliances generally accumulate faults that lead to their eventual demise. However, most researchers do not find this analogy satisfactory as an explanation.

Which of the following would, if true, provide the strongest explanation for the researchers' reaction?

A. Some organisms are capable of living much longer than other organisms.

B. Some organisms reproduce very quickly despite having short lifespans.

C. There are several ways of defining "extreme longevity," and according to some definitions it occurs frequently.

D. Organisms are capable of maintenance and self-repair and can remedy much of the damage that they accumulate.

E. Some organisms generate much more wear and tear on their bodies than others.

Argument Construction

Situation One possible explanation for the fact that natural selection does not favor extreme longevity, despite the fact that a slowed-down aging process would leave more offspring, is that aging is the product of wear and tear, much like how household appliances eventually no longer function because of wear and tear; most researchers, however, reject this analogy as an explanation.

Reasoning *What would explain the researchers' rejection of the explanation?* The explanation is based on analogy between organisms and household appliances. If there were a crucial difference between appliances and organisms, such as the fact that organisms are capable of self-repair in a way that household appliances are not, then it would be reasonable for the researchers to reject the explanation.

A. The fact that some organisms can live much longer than other organisms is not a good reason to reject the analogy; after all, some household appliances may last much longer than others, perhaps because they receive a different amount of wear and tear.

B. Even if some organisms reproduce very quickly despite having short lifespans, that does not mean that they would not have even more offspring if they lived longer; also, this is irrelevant as to whether the analogy between the aging process and the wear and tear on household appliances is any good.

C. It is unclear what these different ways of defining "extreme longevity" are—perhaps the definition is merely a relative one; some organisms live much longer than others, and those that do have "extreme longevity." But presumably, some household appliances last much longer than others, so this would not indicate a difference, and thus would not explain why most researchers reject the analogy.

D. **Correct.** Household appliances cannot repair themselves as organisms can, so the wear and tear of living is quite different from the wear and tear in household appliances.

E. Presumably, some household appliances receive much more wear and tear than other household appliances, so the fact that some organisms generate more wear and tear on their bodies than others does not indicate a difference between appliances and organisms, and thus does not explain why the researchers reject the analogy.

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difference between appliances and organisms, and thus does not explain why the researchers reject the analogy.

The correct answer is D.

•

Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, **any new manufacturing job is created either within an existing company or by the start-up of a new company.** Within existing firms, new jobs have been created this year at well below last year's record pace. At the same time, there is considerable evidence that the number of new companies starting up this year will be no higher than it was last year, and **there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups.** So clearly, the number of new jobs created this year will fall short of last year's record.

In the argument given, the two portions in **boldface** play which of the following roles?

A. The first provides evidence in support of the main conclusion of the argument; the second is a claim that the argument challenges.

B. The first is a generalization that the argument seeks to establish; the second is a conclusion that the argument draws in order to support that generalization.

C. The first is a generalization that the argument seeks to establish; the second is a judgment that has been advanced in order to challenge that generalization.

D. The first is presented as an obvious truth on which the argument is based; the second is a claim that has been advanced in support of a position that the argument opposes.

E. The first is presented as an obvious truth on which the argument is based; the second is a judgment advanced in support of the main conclusion of the argument.

Argument Construction

Situation Manufacturing jobs are created either within existing companies or in start-ups. Manufacturing jobs are being created at a much slower rate this year than last year. It seems likely that the number of new start-ups will not exceed last year's number and that the average number of manufacturing jobs per start-up will not exceed last year's number. So, fewer manufacturing jobs are likely to be created this year than last year.

Reasoning *What function is served by the statement that any new manufacturing job is created either within an existing company or by the start-up of a new company? What function is served by the statement that there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups?* The first statement makes explicit a general background assumption that there are just two ways in which manufacturing jobs are created. This assumption is used, along with other information, to support the argument's main conclusion, i.e., the prediction about job creation this year. The second statement gives a premise meant to help support the prediction about this year's manufacturing-job creation.

A. The first is a general statement making explicit an assumption on which the argument's reasoning depends, but the second is a statement affirmed as part of the argument and does not express a claim that the argument challenges.

B. The first is a generalization that is simply stated, without any support being offered. The second is not a conclusion and is not offered in support of the first.

C. The second is not presented as a challenge to the generalization that is given in the first statement. The argument does not seek to establish the first statement but merely asserts it.

D. The second is information offered in support of the argument's main conclusion rather than a statement offered in support of a position opposed by the argument.

E. **Correct.** The first, stating a truism, is merely asserted and requires no support in the argument, for which it provides a foundation; the second is a piece of information meant to support the prediction that is the argument's main conclusion.

The correct answer is E.

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Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, **any new manufacturing job is created either within an existing company or by the start-up of a new company.** Within existing firms, new jobs have been created this year at well below last year's record pace. At the same time, there is considerable evidence that the number of new companies starting up will be no higher this year than it was last year, and there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups. So clearly, **the number of new jobs created this year will fall short of last year's record.**

In the argument given, the two portions in **boldface** play which of the following roles?

A. The first is presented as an obvious truth on which the argument is based; the second is the main conclusion of the argument.

B. The first is presented as an obvious truth on which the argument is based; the second is a conclusion drawn in order to support the main conclusion of the argument.

C. The first and the second each provide evidence in support of the main conclusion of the argument.

D. The first is a generalization that the argument seeks to establish; the second is the main conclusion of the argument.

E. The first is a generalization that the argument seeks to establish; the second is a conclusion that has been drawn in order to challenge that generalization.

Argument Construction

Situation Manufacturing jobs are created either within existing companies or in start-ups. Manufacturing jobs at existing firms are being created at a much slower rate this year than last year. It seems likely that the number of new start-ups will not exceed last year's number and that the average number of new manufacturing jobs per start-up will not exceed last year's number. So, fewer manufacturing jobs are likely to be created this year than last year.

Reasoning *What function is served by the statement that any new manufacturing job is created either within an existing company or by the start-up of a new company? What function is served by the statement that the number of new jobs created this year will fall short of last year's record number?* The first statement makes explicit a general background assumption that manufacturing jobs are created in just two ways. This assumption is used, along with other information, to support the argument's main conclusion. The second statement gives the argument's main conclusion, a prediction about how this year's manufacturing-job creation will compare with last year's.

A. **Correct.** The first statement expresses a truism: assuming that all manufacturing jobs are created by companies, it is obviously true that all such jobs are created either by companies that already existed or by new companies that did not exist before. The argument then goes on to claim that new-job creation in each of these categories in the current year will be less than in the previous year. Since these two categories are exhaustive (as indicated in the first **boldfaced** portion), the argument concludes that new-job creation in the current year will be less than in the previous year. This is the argument's main conclusion (expressed in the second **boldfaced** portion).

B. The second statement is the argument's main conclusion, not an intermediate conclusion used to support the argument's main conclusion.

C. The second statement is the main conclusion of the argument, not a statement used as support for the main conclusion.

D. The argument merely asserts, and does not "seek to establish," the first statement. The first statement is a truism that does not need to be supported with evidence.

E. The second statement is the argument's main conclusion and is not meant to present a challenge to the first statement. The first statement serves to provide partial support for the argument's main conclusion.

The correct answer is A.

•

In Stenland, many workers have been complaining that they cannot survive on minimum wage, the lowest wage an employer is permitted to pay. The government is proposing to raise the minimum wage. Many employers who pay

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employer is permitted to pay. The government is proposing to raise the minimum wage. Many employers who pay their workers the current minimum wage argue that if it is raised, unemployment will increase because they will no longer be able to afford to employ as many workers.

Which of the following, if true in Stenland, most strongly supports the claim that raising the minimum wage there will not have the effects that the employers predict?

A. For any position with wages below a living wage, the difficulty of finding and retaining employees adds as much to employment costs as would raising wages.

B. Raising the minimum wage does not also increase the amount employers have to contribute in employee benefits.

C. When inflation is taken into account, the proposed new minimum wage is not as high as the current one was when it was introduced.

D. Many employees currently being paid wages at the level of the proposed new minimum wage will demand significant wage increases.

E. Many employers who pay some workers only the minimum wage also pay other workers wages that are much higher than the minimum.

Argument Evaluation

Situation Stenland's government proposes to raise the minimum wage because many workers have complained they cannot survive on it. But many employers claim that raising the minimum wage will increase unemployment.

Reasoning *What evidence would most strongly suggest that raising the minimum wage will not increase unemployment?* The employers with minimum-wage workers implicitly reason that because raising the minimum wage will increase the wages they have to pay each worker, it will reduce the number of workers they can afford to employ, and thus will increase unemployment. Evidence that the increased wage would not actually increase the employers' expenses per employee would cast doubt on their prediction, as would evidence that reducing the number of minimum-wage workers would not increase the nation's overall unemployment rate.

A. **Correct.** This suggests that raising the minimum wage would make it easier for employers to find and retain minimum-wage employees, and that the savings would fully offset the cost of paying the higher wages. If there were such offsetting savings, the employers should still be able to afford to employ as many workers as they currently do.

B. Even if raising the minimum wage does not increase employers' costs for employee benefits, paying the higher wage might still in itself substantially increase employers' overall costs per employee.

C. For all we know, the current minimum wage might have substantially increased unemployment when it was introduced.

D. These additional demands would probably raise employers' overall costs per employee, making it more likely that increasing the minimum wage would increase overall unemployment.

E. Even if some workers receive more than the minimum wage, raising that wage could still raise employers' expenses for employing low-wage workers, making it too expensive for the employers to employ as many workers overall.

The correct answer is A.

•

Biologists with a predilection for theory have tried—and largely failed—to define what it is that makes something a living thing. Organisms take in energy-providing materials and excrete waste products, but so do automobiles. Living things replicate and take part in evolution, but so do some computer programs. We must be open to the possibility that there are living things on other planets. Therefore, we will not be successful in defining what it is that makes something a living thing merely by examining living things on Earth—the only ones we know. Trying to do so is analogous to trying to specify _____.

Which of the following most logically completes the passage?

A. the laws of physics by using pure mathematics

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A. the laws of physics by using pure mathematics

B. what a fish is by listing its chemical components

C. what an animal is by examining a plant

D. what a machine is by examining a sketch of it

E. what a mammal is by examining a zebra

Argument Construction

Situation Some biologists have tried, unsuccessfully, to find a theoretically defensible account of what it means for something to be a living thing. Some of the suggested definitions are too broad because they include things that we would not regard as living. To find life on other planets, we must not narrow our conception of life by basing it simply on the kinds of life encountered on Earth.

Reasoning *Which of the answer choices would be the logically most appropriate completion of the argument?* The argument points out that life-forms elsewhere in the universe may be very different from any of the life-forms on Earth. Both life-forms on Earth and life-forms discovered elsewhere would all qualify as members of a very large class, the class of all life-forms. Taking life-forms on Earth, a mere subset of the class of all life-forms, as representative of all life-forms would be a logical mistake and might not lead to success in defining what it means for something to be a living thing. The correct answer choice, therefore, would involve a case of specifying what some general class of things is by examining the members of only a small, and not necessarily representative, subset of that class of things. In other words, the correct answer choice will involve the logical mistake of taking a subset as representative of a larger class.

A. Pure mathematics is not a subset of the law of physics, so this does not involve the logical mistake of taking a subset as representative of a larger class.

B. The chemical components of a fish are what make up the fish; they are not a small subset of the fish, so this does not involve the logical mistake of taking a subset as representative of a larger class.

C. Plants are not a subset of the class of animals, so this does not involve the logical mistake of taking a subset as representative of a larger class.

D. A sketch of a machine is not a subclass of the machine itself, so this does not involve the logical mistake of taking a subset as representative of a larger class.

E. **Correct.** This involves the logical mistake of taking the class of zebras, a subclass of the class of mammals, as representative of the class of all mammals. Logically, it resembles taking the class of life-forms on Earth as representative of the class of all life-forms.

The correct answer is E.

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